Algorithm Design Paradigms

Handout on Induction

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The rise of nonchalance is the death of passion

— *Anonymous*

§1 Induction

Principle of mathematical induction 1.1. Suppose we have an infinite sequence of statements $\varphi_0, \varphi_1, \varphi_2, \varphi_3, \dots$
and we can prove the following 2 statements -

- φ_0 is true
- For each $n > 0$, if φ_{n-1} is true, then φ_n is also true.

then all the statements $\varphi_0, \varphi_1, \varphi_2, \varphi_3, \dots$ in the sequence are true.

The above definition should be read as follows, given a sequence of formulas:

- The first one is true.
- Any formula being true, implies that the next one in the sequence is true.

Then all of the formulas in the sequence are true. Something like a chain of dominoes falling.

Exercise

Show that n^2 is the same as the sum of first n odd numbers using induction.

The scenic way

(a) Prove the following theorem of Nicomachus by induction:

$$\begin{aligned} 1^3 &= 1 \\ 2^3 &= 3 + 5 \\ 3^3 &= 7 + 9 + 11 \\ 4^3 &= 13 + 15 + 17 + 19 \\ &\vdots \end{aligned}$$

(b) Use this result to prove the remarkable formula

$$1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2$$

Same Height?

Here is a proof by induction that all people have the same height. We prove that for any positive integer n, any group of n people all have the same height. This is clearly true for $n = 1$. Now assume it for n, and suppose we have a group of $n + 1$ persons, say $P_1, P_2, ..., P_{n+1}$. By the induction hypothesis, the n people $P_1, P_2, ..., P_n$ all have the same height. Similarly the n people $P_2, P_3, ..., P_{n+1}$ all have the same height. Both groups of people contain $P_2, P_3, ..., P_n$, so P_1 and P_{n+1} have the same height as $P_2, P_3, ..., P_n$. Thus all of $P_1, P_2, ..., P_{n+1}$ have the same height. Hence by induction, for any n any group of n people have the same height. Letting n be the total number of people in the world, we conclude that all people have the same height. Is there a flaw in this argument?

There is enough information!

Given $a_0 = 100$ and $a_n = -a_{n-1} - a_{n-2}$, what is a_{2025}?

2-3 Color Theorem

A k-coloring is said to exist if the regions the plane is divided off in can be colored with k colors in such a way that no two regions sharing some length of border are the same color.

(a) A finite number of circles (possibly intersecting and touching) are drawn on a paper. Prove that a valid 2-coloring of the regions divided off by the circles exists.

(b) A circle and a chord of that circle are drawn in a plane. Then a second circle and chord of that circle are added. Repeating this process, until there are n circles with chords drawn, prove that a valid 3-coloring of the regions in the plane divided off by the circles and chords exists.

Square-full

Call an integer square-full if each of its prime factors occurs to a second power (at least). Prove that there are infinitely many pairs of consecutive square-fulls.

Hint: We recommend using induction. Given $(a, a + 1)$ are square-full, can we generate another?

Proving the principle of induction

Prove that the following statements are equivalent -

- every nonempty subset of $\mathbb{N}$ has a smallest element
- the definition of induction

You can assume that $<$ is a linear order on $\mathbb{N}$ with $n - 1 < n$
and such that there are no elements strictly between $n - 1$ and n.